Microsoft Certified Azure Administrator: AZ-104

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TECHNOLOGY

Deploy and Manage Azure Compute Resources

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A Day in the Life of an Azure Architect

As an Azure Administrator responsible for managing the servers that support the company's entire infrastructure, from the web to database servers, your role includes accountability for deploying new virtual machines. The preparation tasks can embrace properly filling the machines, choosing storage, and configuring networking.

It is often difficult to manage virtual machines at scale, particularly when usage patterns vary and demands on applications fluctuate. Users would like the ability to adjust virtual machine resources to match these demands while also maintaining consistent configurations to ensure application stability.

Users will be accountable to manage the platform as a service choice, such as Azure app service and Azure Kubernetes service.

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Learning Objectives

By the end of this lesson, you will be able to:

- Create virtual machines with operating system, storage, and networking capabilities
- Define an app service to configure one or more apps to run in the same App Service plan
- Generalize Azure container instances for the fastest and simplest way to run containers in Azure
- Apply Azure disk encryption to safeguard and preserve the data

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Administer Virtual Machines

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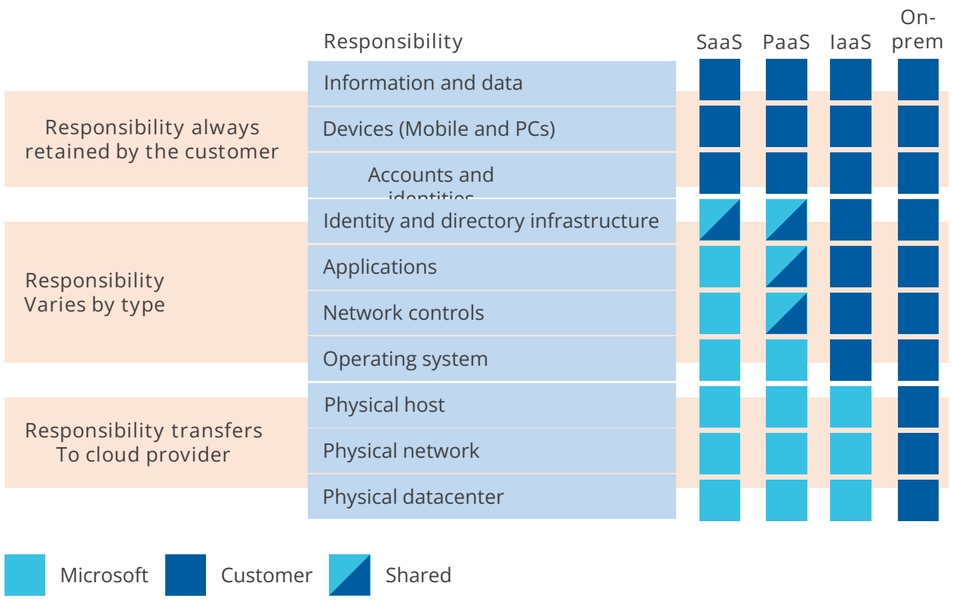
Cloud Services Responsibilities

Virtual machines are a crucial part of Infrastructure as a Service (IaaS) providing. IaaS provides a fast, internet-based computing infrastructure that allows users to quickly scale up or down based on demand and pay only for the resources they use.



IaaS business scenarios

Test and development

- Website hosting
- Storage, backup, and recovery
- High-performance computing
- Big data analysis
- Extended data center

Source: https://learn.microsoft.com/en-us/azure/security/fundamentals/shared-responsibility

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Plan Virtual Machines

Azure virtual machines give an operating system, storage, and networking capabilities, allowing them to run a wide range of applications.

- Start with the network
- Name the virtual machine

Choose a location:

- Each region has completely different hardware and service capabilities
- Locate virtual machines as near as possible to the users to ensure compliance and legal obligations

Consider pricing:

- Compute prices
- Storage prices (consumption-based and reserved instances)

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Moving a Resource to Another Resource Group

Duration: 10 Min.

Problem Statement:

You've been tasked to move a resource to another resource group.

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Assisted Practice: Guidelines

Steps to be followed:

1. Move a resource to another resource group

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Determine Virtual Machine Sizing

Azure offers users VM size choices with different combinations of workout, memory, and storage.

Type	Description
General purpose	Balanced CPU-to-memory ratio. Ideal for testing and development, small to medium databases, and low to medium traffic web servers.
Compute-optimized	High CPU-to-memory ratio. Good for medium traffic web servers, network appliances, batch processes, and application servers.
Memory optimized	High memory-to-CPU ratio. Great for relational database servers, medium to large caches, and in-memory analytics.

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Determine Virtual Machine Sizing

Type	Description
Storage optimized	High disk throughput and IO ideal for Big Data, SQL, NoSQL databases, data warehousing, and large transactional databases.
GPU	Specialized virtual machines targeted for heavy graphic rendering and video editing, as well as model training and inferencing (ND) with deep learning. Available with single or multiple GPUs.
High performance compute	The fastest and most powerful CPU virtual machines with optional high-throughput network interfaces (RDMA).

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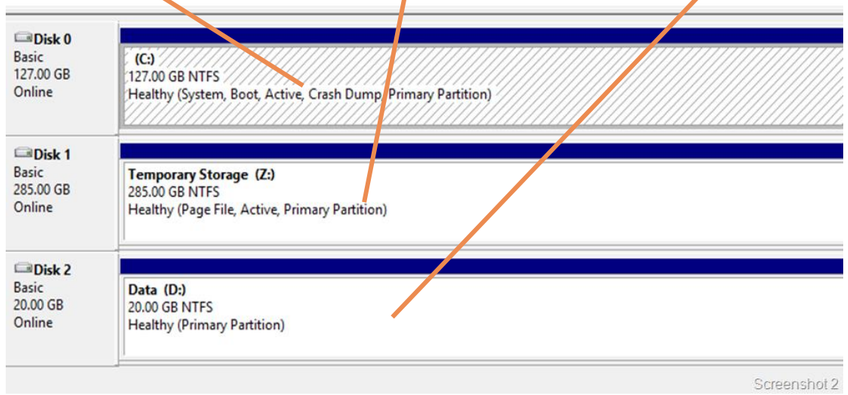
Determine Virtual Machine Storage

OS Disk
- Drive C:
- persistent

Temporary Disk
- Drive D:
- Local (Not persistent)

Data Disk(s)
- Customer Defined Letter
- persistent



Each Azure VM has two or more disks:
- OS disk
- Temporary disk (contents can be lost)
- Data disks (optional)

When creating an Azure VM, users can select between:
- Managed disks (recommended)
- Unmanaged disks

OS and data disks reside in Azure Storage accounts:
- Azure-based storage service
- Standard (HDD, SSD) or Premium (SSD), or Ultra (SSD)

Source: https://learn.microsoft.com/en-us/azure/virtual-machines/managed-disks-overview

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Creating Virtual Machine from Azure Portal

Duration: 10 Min.

Problem Statement:

You've been tasked to create and configure a virtual machine using the Azure portal.

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Assisted Practice: Guidelines

Steps to be followed:

1. Go to the Azure portal
2. Locate the virtual machine
3. Create a virtual machine

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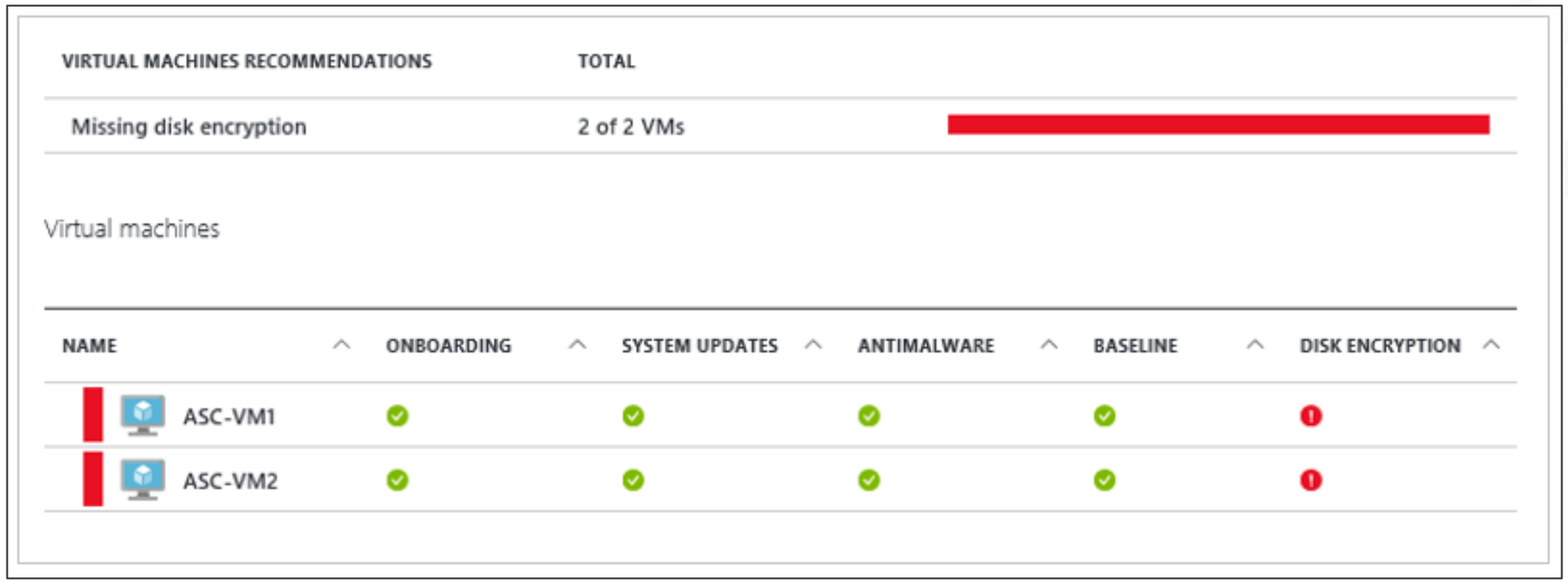
Configure Azure Disk Encryption

- Azure disk encryption is zone-resilient like VMs
- Microsoft Defender for cloud issues alerts for unencrypted VMs, marked as high severity, with recommendations to enable encryption



VIRTUAL MACHINES RECOMMENDATIONS	TOTAL
Missing disk encryption	2 of 2 VMs

Virtual machines

NAME	ONBOARDING	SYSTEM UPDATES	ANTIMALWARE	BASELINE	DISK ENCRYPTION
ASC-VM1	✓	✓	✓	✓	!
ASC-VM2	✓	✓	✓	✓	!

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Configure Azure Disk Encryption

- Azure disk encryption safeguards and preserves data
- It uses Windows BitLocker for volume encryption on both the operating system and data drives of Azure virtual machines (VMs)
- It seamlessly integrates with Azure key vault
- It enables efficient management of encryption keys and confidential information associated with the drives

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Supported VMs and Operating Systems

Supported VMs

- Linux VMs are available in various series
- Azure Disk Encryption is compatible with Generation 1 and Generation 2 VMs and is also accessible for VMs utilizing premium storage
- It is unavailable for Basic and A-series VMs, as well as for virtual machines that do not meet the specified minimum memory requirements

Supported operating systems

- Supported OS range: Windows 8 through Windows 10 Enterprise multi-session
- Specified key size requirements for Windows Server 2022 and Windows 11

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Configure Azure Disk Encryption: Prerequisites

Disk encryption can only be applied to virtual machines that meet the following prerequisites:

- Networking requirements
- Group policy requirement
- Encryption key storage requirement

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Configure Azure Disk Encryption: Prerequisites

Networking requirements

The Windows VM must connect to an Azure active directory endpoint to acquire a token for key vault access.

Group policy requirement

The Azure VMs' Internet access is restricted, so users can address the URI resolution issue by establishing a custom rule to allow outbound connectivity to the required IP addresses.

Encryption key storage requirements

The Azure disk encryption necessitates the use of an Azure Key Vault for the management and storage of disk encryption keys and secrets.

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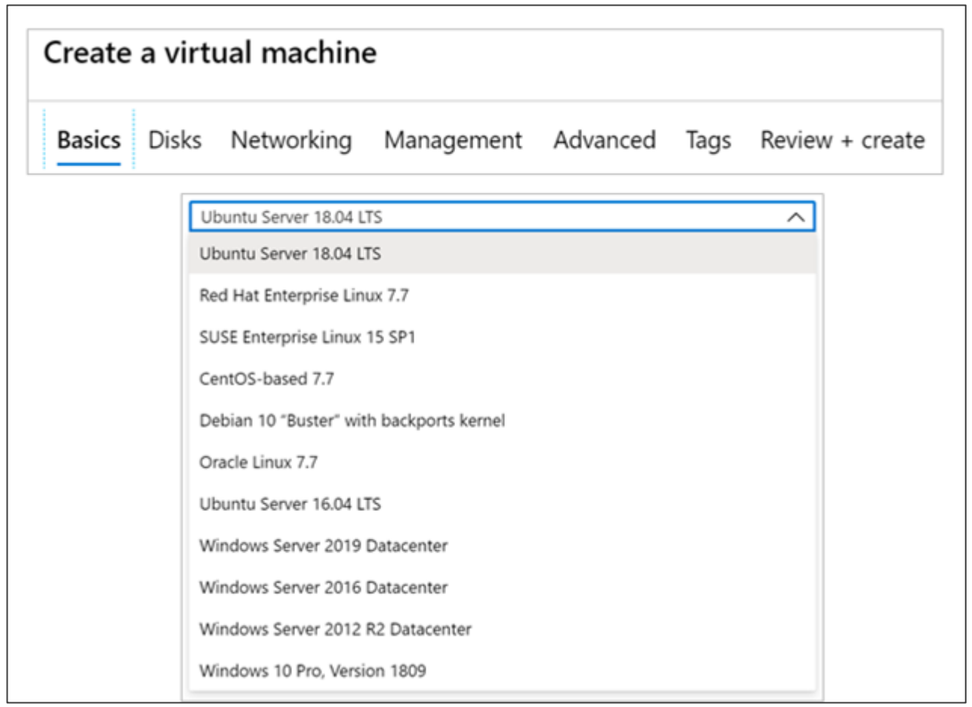
Create Virtual Machines in the Portal

The basic requirements to create a VM in the portal include:



- Basic (required): Project details, administrator account, inbound port rules
- Disks: OS disk type, data disks
- Networking: Virtual networks, load balancing
- Management: Monitoring, auto-shutdown, backup
- Advanced: Add additional configuration, agents, scripts, or applications

Source: https://docs.microsoft.com/en-us/azure

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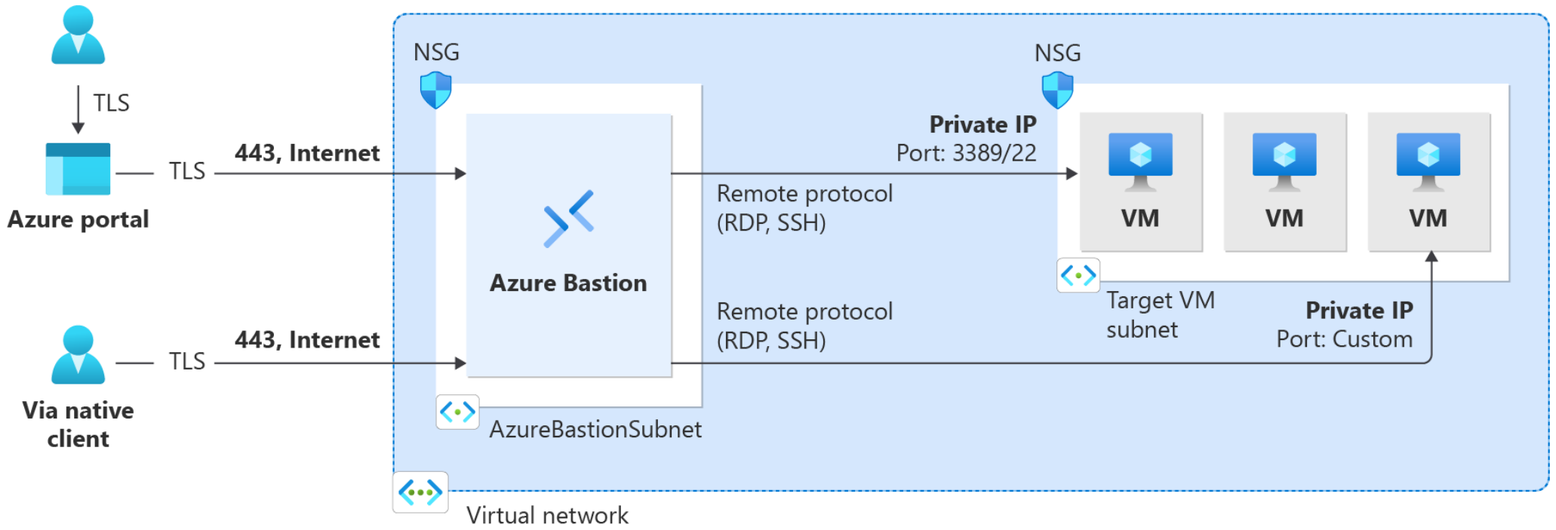
Connect to Virtual Machines

Virtual machines in Azure can be accessed in the following ways: :

- Bastion subnet for RDP/SSH through the portal over SSL
- Remote desktop protocol for Windows-based virtual machines
- Secure shell protocol for Linux-based virtual machines



Source: https://learn.microsoft.com/en-us/azure/bastion/bastion-overview

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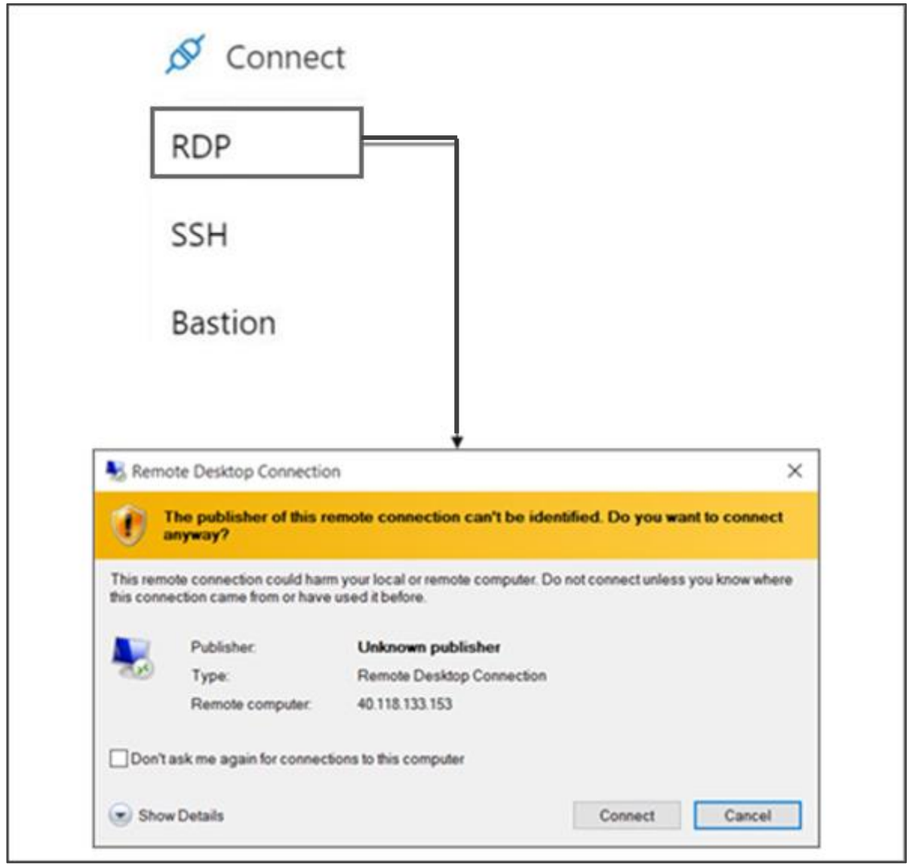
Connect to Windows Virtual Machines

Users may use the following strategies to address unit OS specific problems:



Remote Desktop Protocol (RDP) creates a GUI session and accepts inbound traffic on TCP port 3389

WinRM creates a command-line session for the user to run scripts.

Source: https://docs.microsoft.com/en-us/azure/bastion/bastion-overview

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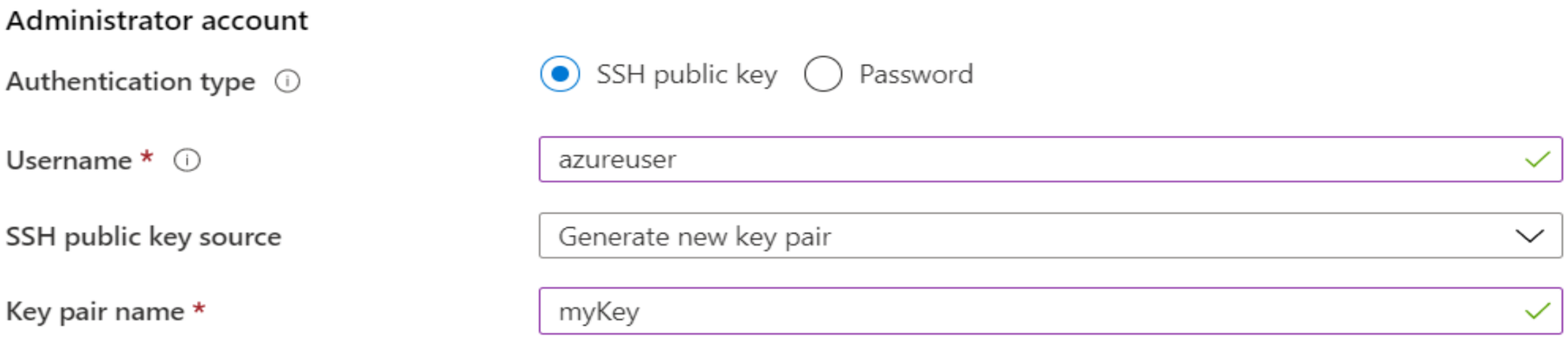
Connect to Linux Virtual Machines

About nursing SSH public key:

- An SSH public key or password is required to authenticate.
- SSH is an encrypted connection protocol that allows secure logins over unsecured connections.
- These are also called public-private keys.



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Connect to a Windows Virtual Machine

Duration: 10 Min.

Problem Statement:

You've been tasked to connect a Windows-based virtual machine using RDP on port 3389.

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Steps to be followed:

1. Go to Virtual machine blade
2. Select the VM
3. Click on Connect
4. Download the RDP file
5. Log in using the admin username and admin password provided during the VM creation

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Connect to a Linux Virtual Machine

Duration: 10 Min.

Problem Statement:

You've been tasked to connect a Linux-based virtual machine using SSH on port 22.

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Steps to be followed:

1. Log in to Cloud Shell
2. Create SSH key pair
3. Use the public key while creating the VM
4. Connect to VM

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Plan for Maintenance and Downtime

Situations impacting the virtual machine in Azure include:

Unplanned hardware maintenance

When the platform predicts a failure, it will issue an unplanned hardware maintenance event.

Action: Live migration

Unexpected downtime

Unexpected downtime is when a virtual machine fails unexpectedly.

Action: Automatically migrate (heal)

Planned maintenance

Planned maintenance events are periodic updates made to the Azure platform.

Action: No action

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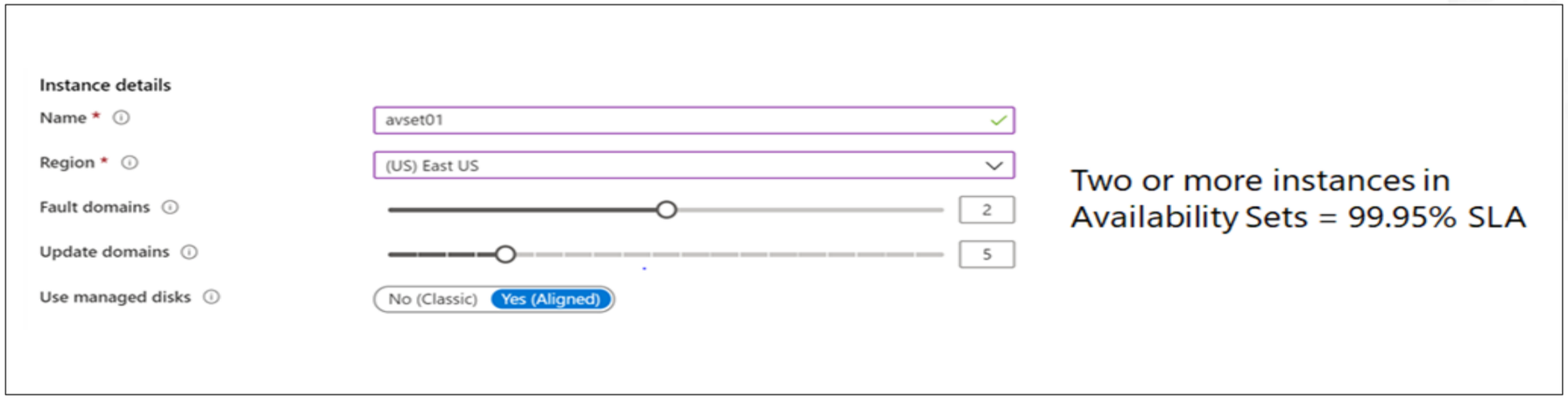
Setup Availability Sets

Availability sets are essential to build reliable cloud solutions. General principles include:

- Configure multiple virtual machines in an availability set
- Configure each application tier into separate availability sets
- Combine a load balancer with availability sets
- Use managed disks with the virtual machines



Two or more instances in Availability Sets = 99.95% SLA

Source: https://learn.microsoft.com/azure/virtual-machines/availability-set-overview

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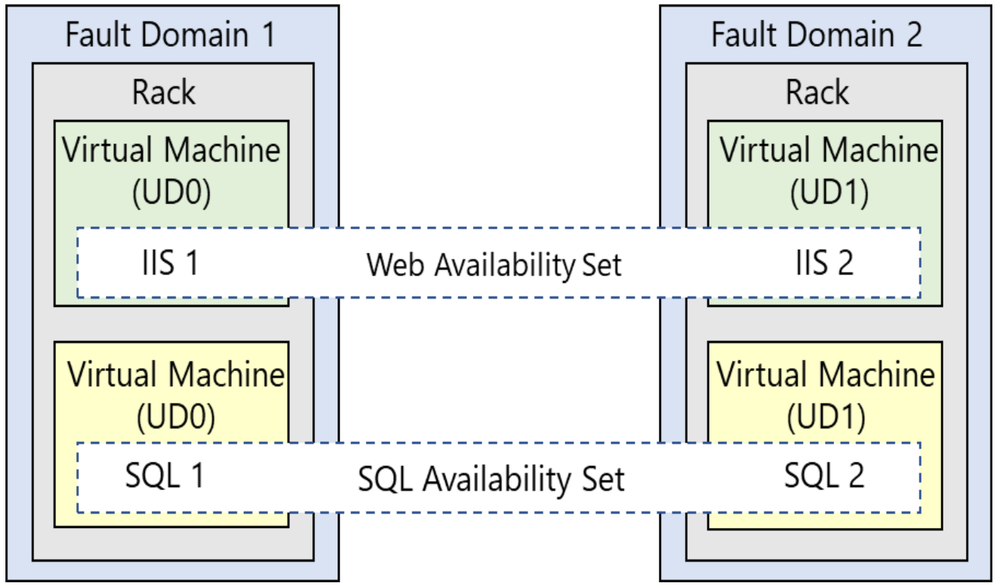
Update and Fault Domains

Each virtual machine in the availability set has an update domain and fault domain assigned to ensure high availability and redundancy.



Update domains allow Azure to perform incremental or rolling upgrades across a deployment. During planned maintenance, only one update domain is rebooted at a time.

Fault domains are a group of virtual machines that share the same hardware and switches that share a single point of failure. VMs in an availability set are placed in at least two fault domains.

Source: https://docs.microsoft.com/en-us/azure

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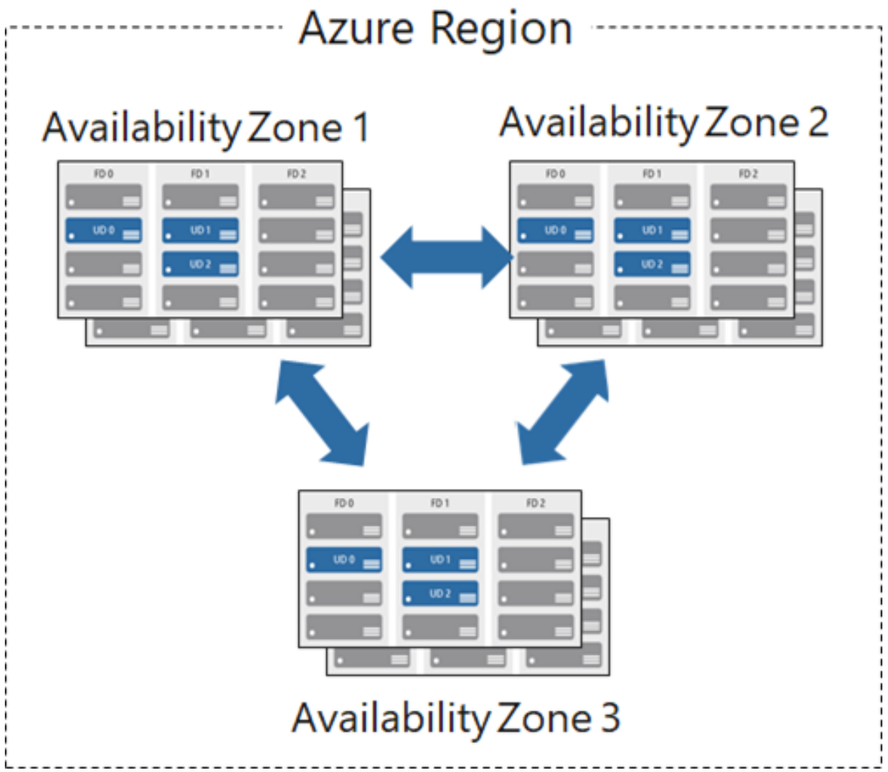
Availability Zones

Considerations for availability zones to protect applications and data from datacenter failures include:



- Unique physical locations in a region
- Includes data centers with independent power, cooling, and networking
- Protects from data center failures
- Combines update and fault domains
- Provides 99.99% SLA

Source: https://docs.microsoft.com/en-us/azure

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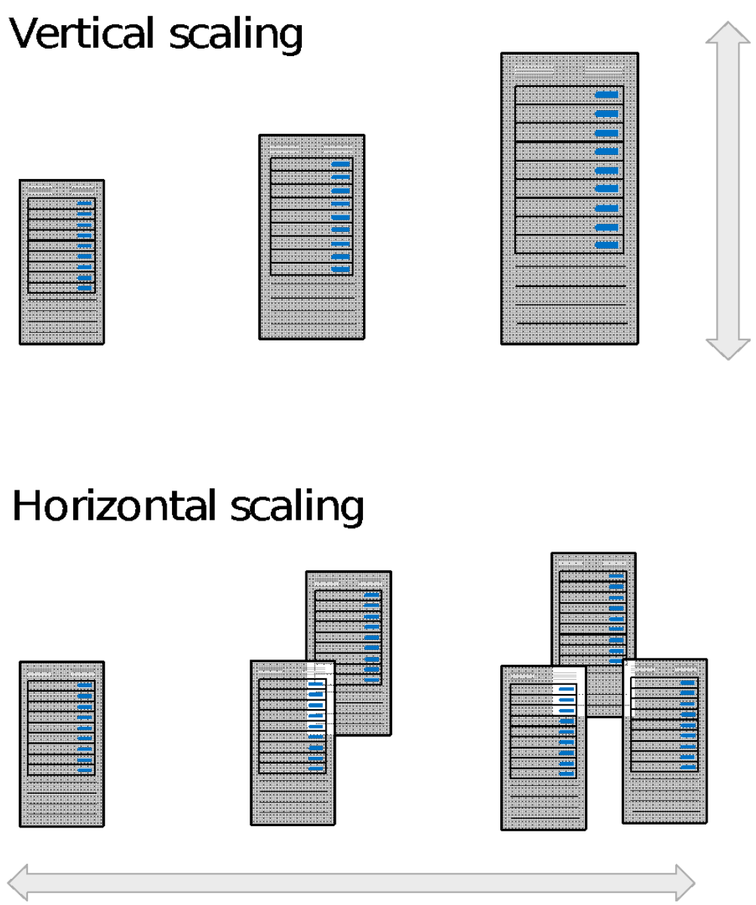
Vertical Scaling Vs. Horizontal Scaling



- Vertical scaling (scale up and scale down) is the process of increasing or decreasing power to a single instance of a workload, usually manual.
- Horizontal scaling (scale out and scale in) is the process of increasing or decreasing the number of instances of a workload. It is frequently automated.

Source: https://docs.microsoft.com/en-us/azure

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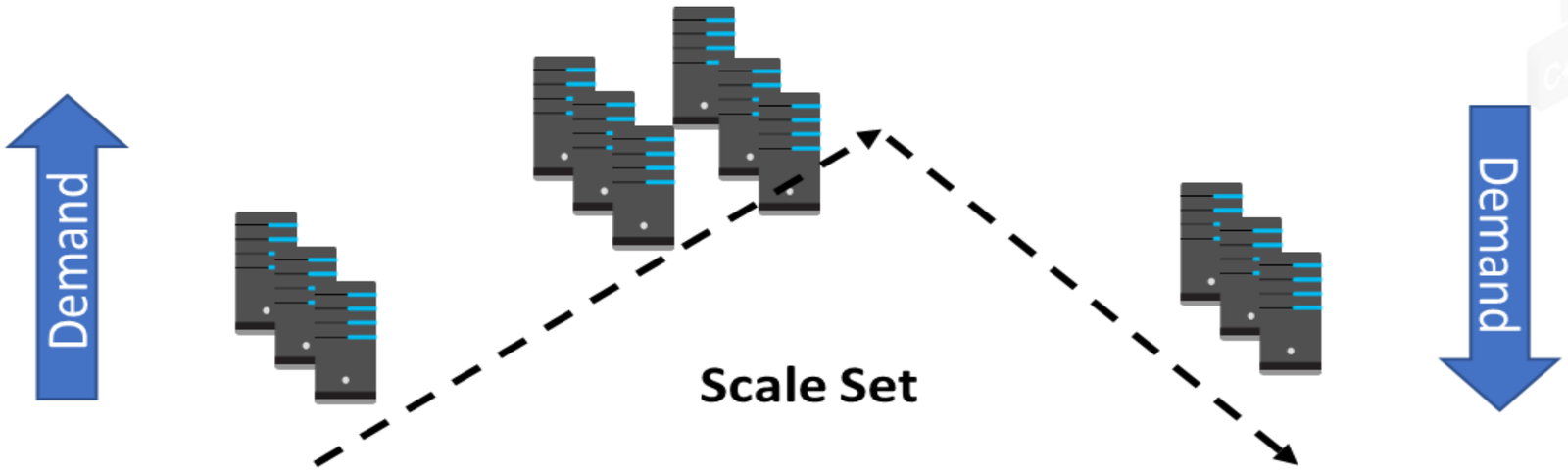
Implement Scale Sets

The benefits of the Scale set are:

- They deploy a set of identical VMs
- No pre-provisioning of VMs is required
- As demand goes up, VMs are added
- As demand goes down, VMs are removed
- The process can be manual, automated, or a combination of both



Source: https://docs.microsoft.com/en-us/azure

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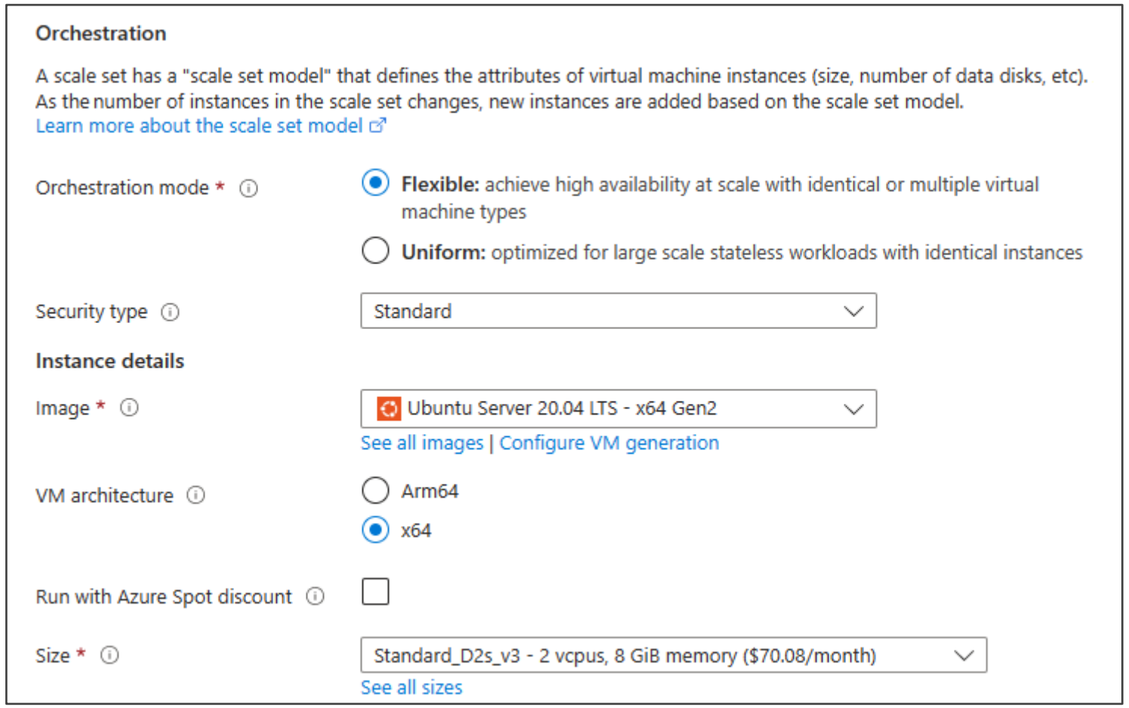
Create Scale Sets



Consider the following parameters when creating a scale set:

- Instance count: Number of VMs in the scale set (0 to 1000)
- Instance size: The size of each virtual machine in the scale set
- Azure Spot instance: Unused capacity at a discounted rate
- Use managed disks
- Enable scaling beyond 100 instances

Source: https://learn.microsoft.com/en-us/azure/virtual-machine-scale-sets/flexible-virtual-machine-scale-sets-portal

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Create Scale Sets

Duration: 10 Min.

Problem Statement:

You’ve been tasked to create a virtual machine scale set that will be flexible enough to adjust the instance count based on the load.

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Assisted Practice: Guidelines

Steps to be followed:

1. Locate the virtual machine scale set service in the Azure portal
2. Select create
3. Provide the necessary details to create the scale set

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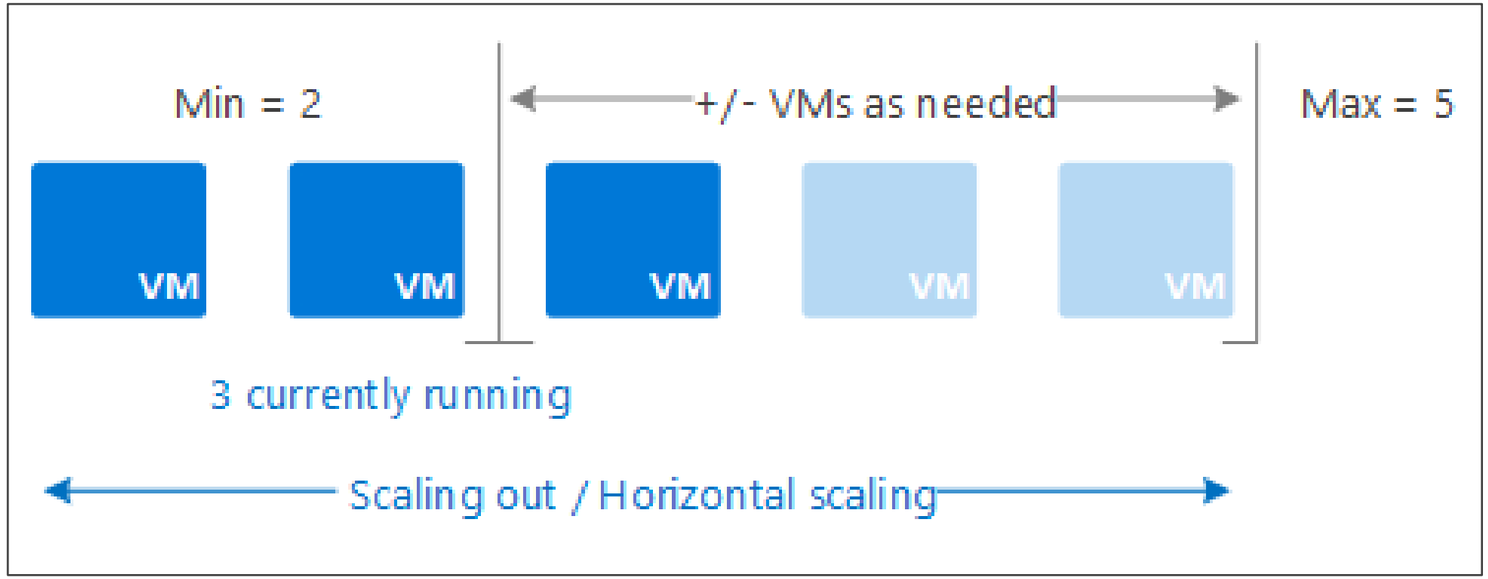
Implement Autoscale

Autoscale benefits:

- Defines rules to automatically adjust capacity
- Scales out (increase) the number of VMs in the set
- Scales in (reduce) the number of VMs in the set
- Schedules events to increase or decrease at a fixed time
- Reduces monitoring and optimizes performance



Source: https://learn.microsoft.com/en-us/azure/azure-monitor/autoscale/autoscale-overview

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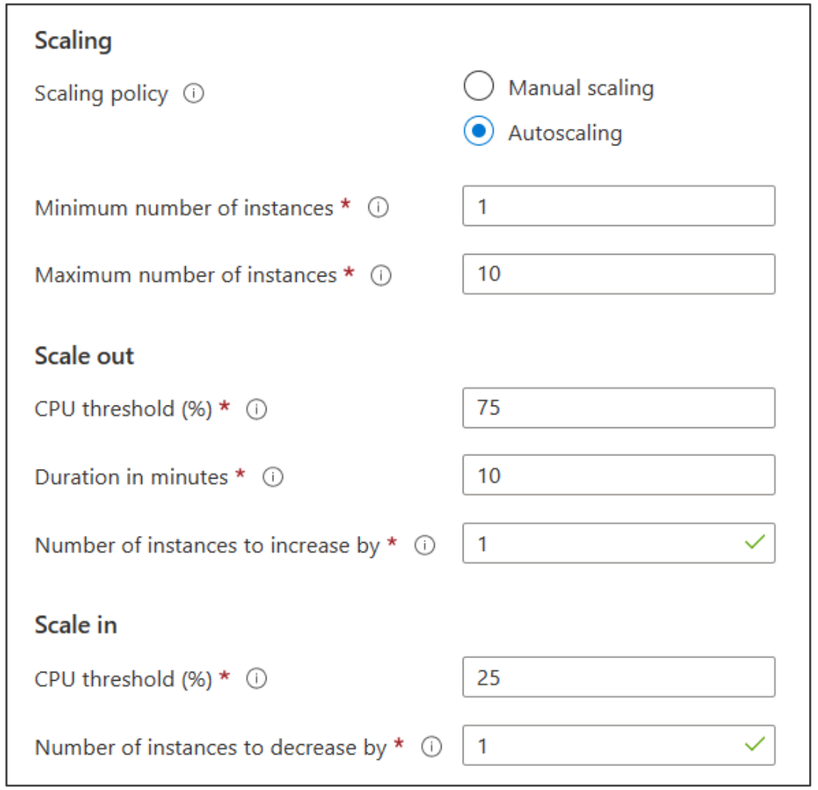
Configure Autoscale

Autoscale can be enabled while creating a scale set, along with defining a minimum, maximum, and default number of VM instances.



- Define a minimum, maximum, and default number of VM instances
- Create more advanced scale sets with scale out and scale in parameters

Source: https://learn.microsoft.com/en-us/azure/virtual-machine-scale-sets/virtual-machine-scale-sets-autoscale-overview

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Configure Autoscaling

Duration: 05 Min.

Problem Statement:

You've been tasked to configure the autoscaling in the virtual machine scale set so that, based on this setting, the scale set will automatically scale in and scale out.

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Assisted Practice: Guidelines

Steps to be followed :

1. Sign in to the Azure portal
2. Locate the scale set created
3. Choose scaling from the menu to enable autoscaling

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Query System and Runtime Information about VM

Duration: 15 Min.

Problem Statement:

You've been tasked to query the system and runtime information about the VM to get information about the virtual machine.

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Assisted Practice: Guidelines

Steps to be followed:

1. Login to Cloud Shell
2. Use az vm list command to list the VMs defined in the Azure subscription
3. Run az vm show command to fetch the detailed VM information

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Create and Configure an App Service

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Implement Azure App Service Plans

Features of App Service plan:

- Defines a set of computing resources for a web app to run
- Determines performance, price, and features
- Configures one or more apps to run in the same App Service plan
- Segregates regions where compute resources will be created
- Decides the number of virtual machine instances
- Gives the size of virtual machine instances

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Determine App Service Plan Pricing

The pricing tier of an App Service plan decides the features and payment details.

Shared compute
(Free and Shared).
Run app
on the same Azure VM as other app service apps, and the resources cannot scale out.

Dedicated compute
(Basic, Standard, Premium).
Run app in the same plan on dedicated Azure VMs.

Isolated
Runs apps on dedicated Azure VMs in dedicated Azure virtual networks.

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Determine App Service Plan Pricing

Selected Features	Free	Shared (dev/test)	Basic (dedicated dev/test)	Standard (production workloads)	Premium (enhanced scale and performance)	Isolated (high-performance, security and isolation)
Web, mobile, or API apps	10	100	Unlimited	Unlimited	Unlimited	Unlimited
Disk space	1 GB	1 GB	10 GB	50 GB	250 GB	1 TB
Auto Scale	–	–	–	Supported	Supported	Supported
Deployment Slots	0	0	0	5	20	20
Max Instances	–	–	Up to 3	Up to 10	Up to 30	Up to 100

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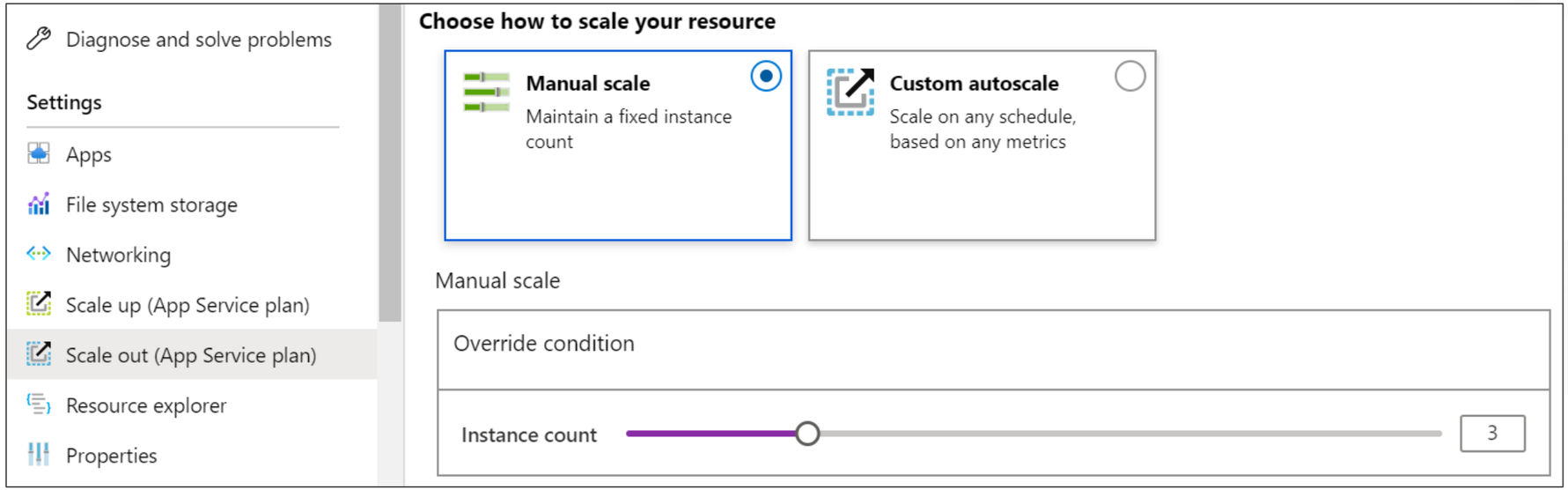
Scale up and Scale out the App Service Plan

Scale up (change the App Service plan):

- More hardware (CPU, memory, disk)
- More features (dedicated virtual machines, staging slots, autoscaling)

Scale-out (increase the number of VM instances):

- Manual (fixed number of instances)
- Auto scale (based on predefined rules and schedules)



Source: https://docs.microsoft.com/en-us/azure

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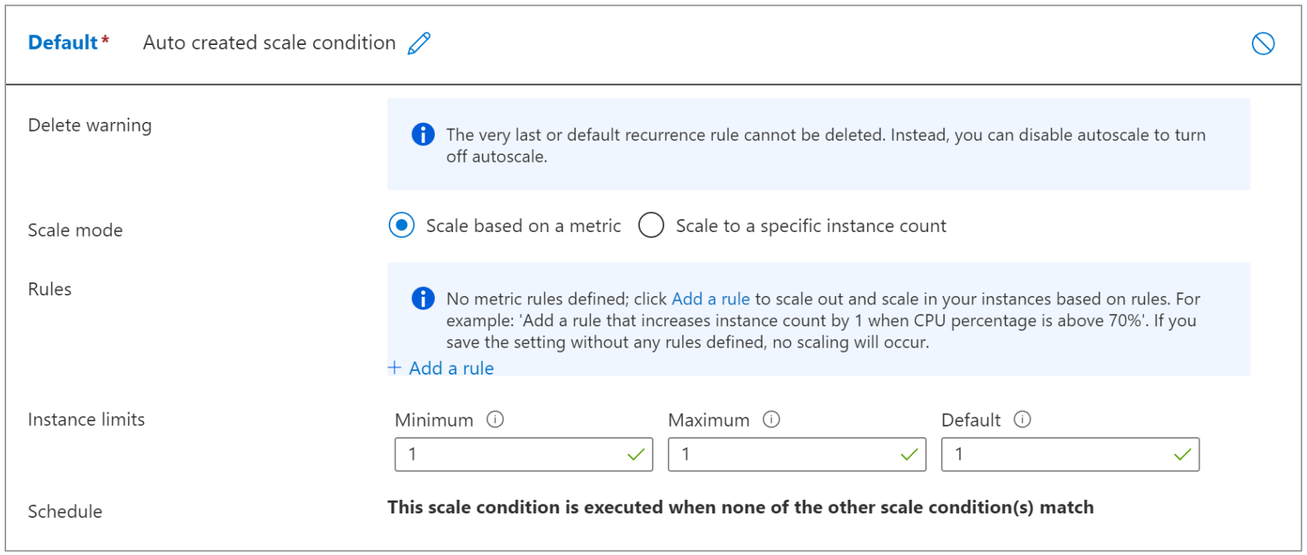
Configure App Service Plan Scaling

Autoscale has the right amount of resources running to handle the load on the application. Below are a few features:

- Adjusts available resources based on the current demand
- Improves availability and fault tolerance
- Scales based on a metric (CPU percentage, memory percentage, HTTP requests)
- Scales according to a schedule (weekdays, weekends, times, holidays)
- Implements multiple rules: Combine metrics and schedules



Source: https://docs.microsoft.com/en-us/azure

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Scale up App Service

Duration: 05 Min.

Problem Statement:

You've been tasked to configure the autoscaling in an App Service so that based on this setting the scale set will automatically scale in and scale out.

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Assisted Practice: Guidelines

Steps to be followed :

1. Sign in to the Azure portal
2. Navigate to the App Service app page
3. Select scale up option

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Implement Azure App Service

Reasons to use App Services:

- Includes web apps, API apps, mobile apps, and function apps
- Manages environment enabling high productivity development
- Platform-as-a-service (PaaS) enables building and deploying highly available cloud apps for web and mobile
- Handles infrastructure so developers can focus on core web apps and services
- Provides developer productivity using .NET, .NET Core, Java, Python, and a host of others
- Provides enterprise grade security and compliance

.NET | Node.js | PHP | Java | Python (on Linux) | HTML | Custom Windows/Linux Container

Source: https://docs.microsoft.com/en-us/azure

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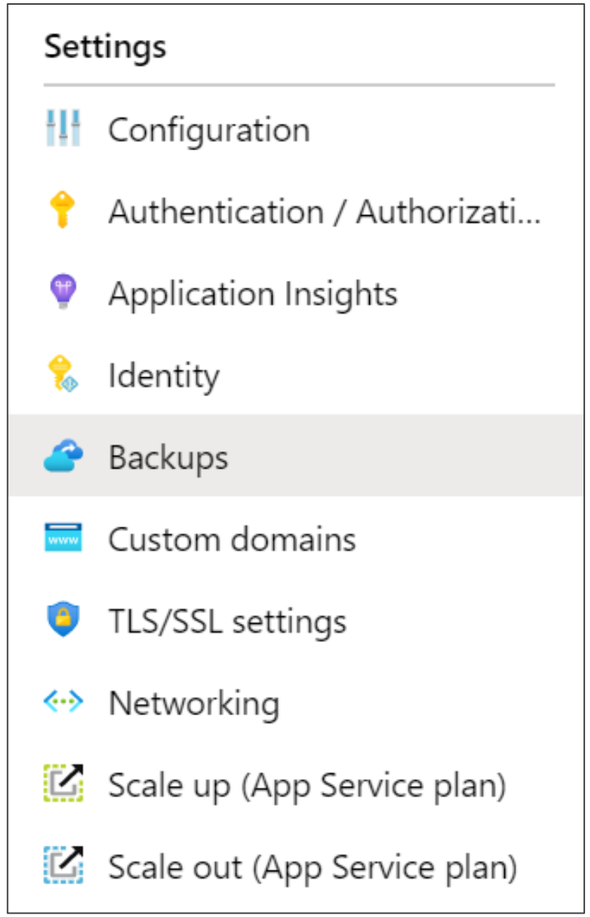
Backup an App Service



Tips to backup an App Service:

- Offers convenient app backup and restore options
- Creates app backups manually or on a schedule
- Backs up the configuration, file content, and database connected to the app
- Requires a standard or premium plan
- Backups can be up to 10 GB of app and database content
- Configures partial backups and exclude items from the backup
- Restores the app on-demand to a previous state, or create a new app

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Creating an App Service Plan

Duration: 10 Min.

Problem Statement:

You've been tasked to create an App Service plan.

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Assisted Practice: Guidelines

Steps to be followed :

1. Sign in to the Azure portal
2. Configure the details required and create the App Service plan
3. Monitor the app service Plan and view all the apps present in the plan

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Deploy Code to App Service

Duration: 10 Min.

Problem Statement:

You've been tasked to deploy the code to the app service and test the application after deployment.

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Assisted Practice: Guidelines

Steps to be followed:

1. Create an App Service plan
2. Create an App Service
3. Deploy the sample application
4. Test the web app

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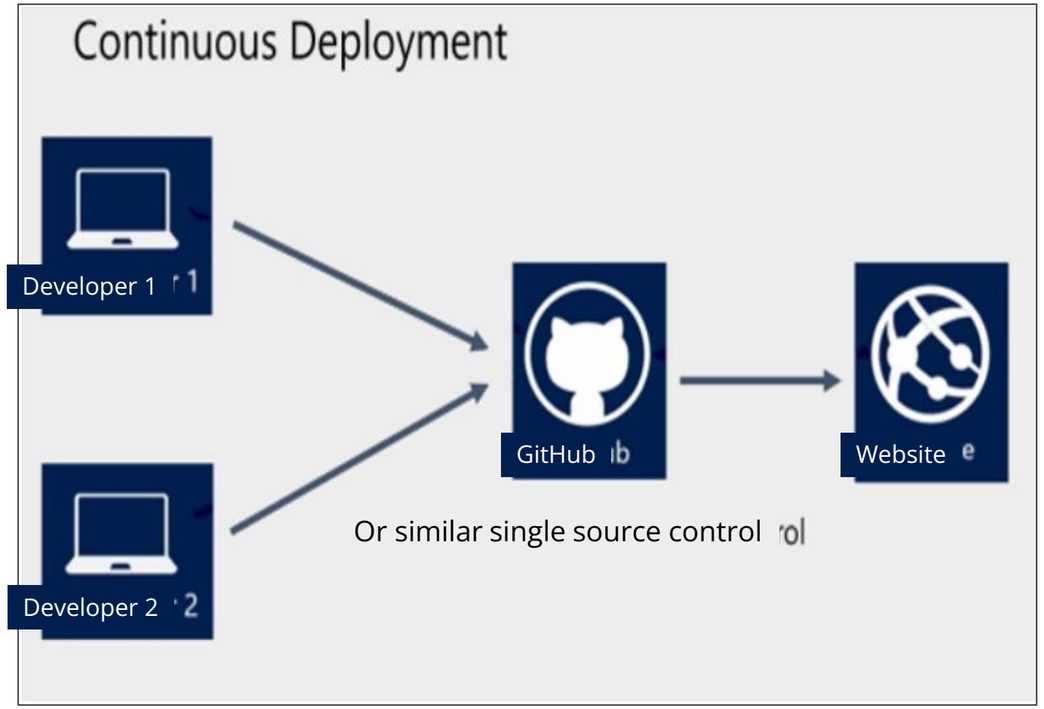
Explore Continuous Integration and Deployment



- It connects the web app with any of the single source control and the App Service auto syncs the code into the web app.
- Whenever code updates are pushed to the source control, the website or web app automatically picks up the updates.
- A continuous deployment workflow publishes the most recent updates from a project.
- Azure portal provides continuous deployments from GitHub, Bitbucket, or Azure DevOps.

Source: https://docs.microsoft.com/en-us/azure

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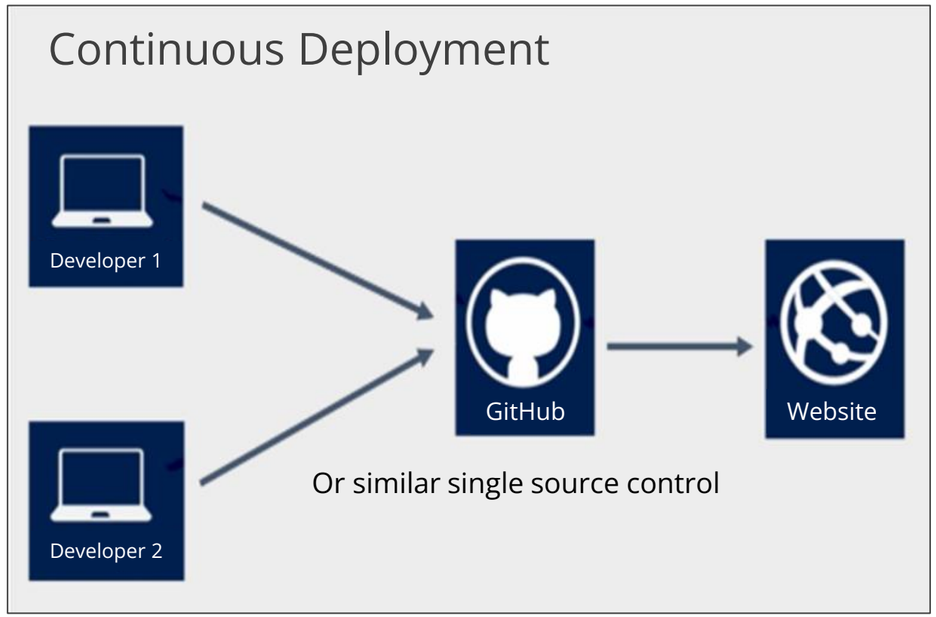
Create Deployment Slots

Service Plan	Slots
Free, Shared, Basic	0
Standard	Up to 5
Premium	Up to 20
Isolated	Up to 20



Using separate staging and production slots has several advantages:

- Deploy to different deployment slots (depending on service plan)
- Validate changes before sending to production
- Deployment slots are live apps with their hostnames
- Avoiding a cold start eliminates downtime
- Fallback to a last known good site
- Auto Swap when pre-swap validation is not needed

Source: https://docs.microsoft.com/en-us/azure

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Add Deployment Slots

Features and types of deployment slots:

- When selecting whether to clone an app configuration from another deployment slot, pay attention to the settings
- Slot-specific app settings and connection strings
- Continuous deployment settings
- App Service authentication settings
- Not all settings are sticky (endpoints, custom domain names, SSL certificates, and scaling)
- Review and edit settings before swapping

Source: https://docs.microsoft.com/en-us/azure

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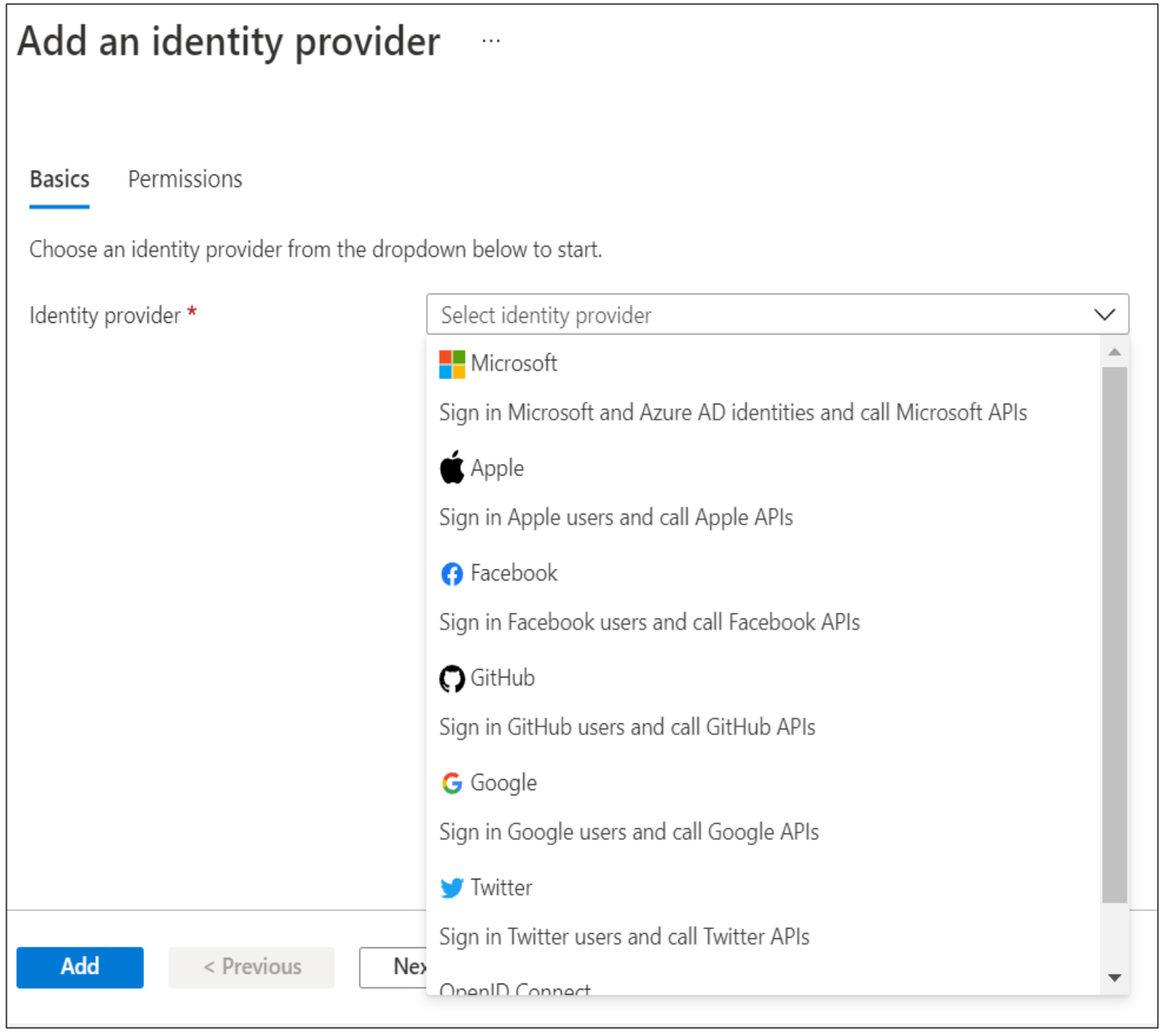
Secure an App Service



Authentication

- Enable authentication: Default anonymous
- Log in with a third-party identity provider

Security

- Troubleshoot with diagnostic logs: Failed requests, app logging
- Add an SSL certificate: HTTPS
- Define a priority ordered allow or deny list to control network access to the app
- Store secrets in the Azure key vault

Source: https://docs.microsoft.com/azure/app-service/security-recommendations

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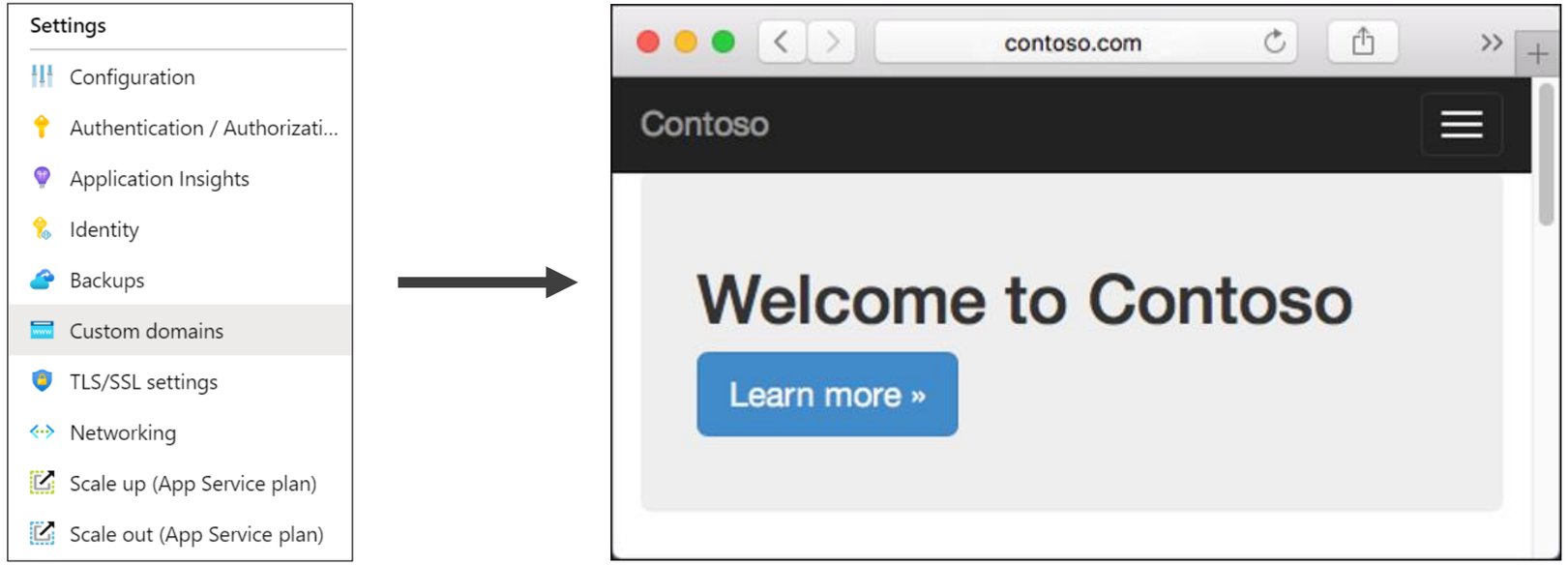
Create Custom Domain Names

To create a web app by setting up a custom domain:

- Redirect the default web app URL
- Validate the custom domain in Azure
- Use the DNS registry for the domain provider: Create a CNAME or A record with the mapping
- Ensure the App Service plan supports custom domains



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Automatics Vs. Custom Backup

Automatic backup

- It occurs regularly in supported pricing tiers
- It does not need configuration
- Its backup size is 30 GB
- It does not need a storage account
- Its retention is 30 days, not configurable
- It is not downloadable
- It does not support partial backup
- It does not support backup over VNet

Custom backup`

- Requires initial setup and can be done on-demand or scheduled.
- It needs configuration
- Its backup size is 10 GB, of which 4 GB can be the linked database
- It needs a storage account
- Its retention is 0-30 days or indefinite
- It is not downloadable (Azure storage blobs)
- It supports partial backup supports back up over VNet

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Restore a Backup

Log in to the Azure portal and select Backups in the left menu. This page shows all the automatic and custom backups and the status of each

Choose any of the backups to restore by clicking on the Restore link

Backup details will open

Choose a destination

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Restore a Backup

Select Create new under the App Service box to restore to a new app and select Create new under the Deployment slot box for restoring to a new deployment slot

Choose to restore the site configuration under the Advanced options

Click on Restore

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How Backups are Stored

- Backups are accessible via the containers tab in the storage account.
- Each backup is a .zip file containing data and an .xml file describing its contents.
- It can unzip and inspect these files to view backups without initiating a restoration process.
- To check for error messages for failed backups, select the corresponding item on the backups page; this is referred to as an error message.

Automate with scripts

Using either the Azure CLI or Azure PowerShell, users can employ scripts for the automated management of backups.

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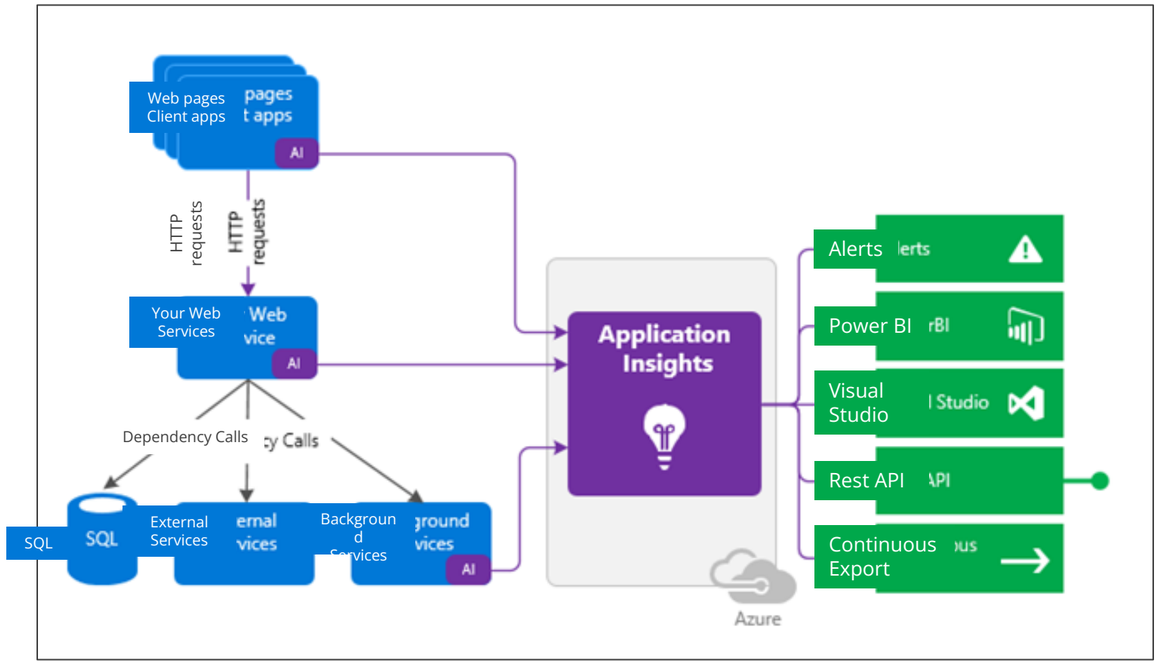
Use Application Insights



Features of application insights:

- Request rates, deny rates, response time, and failure rates
- Page views and load performance
- User and session counts
- Performance counters
- Diagnostics and exceptions

Source: https://docs.microsoft.com/en-us/azure

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Create and Configure Azure Containers and Desired State Configurations

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Compare Containers to Virtual Machines

The following table depicts the differences between containers and virtual machines:

Feature	Containers	Virtual Machines
Isolation	Provide lightweight isolation from the host and other containers but doesn't provide as strong a security boundary as a virtual machine	Provide complete isolation from the host operating system and other VMs. This is useful when a strong security boundary is critical, such as hosting apps from competing companies on the same server or cluster
Operating system	Run the user mode portion of an operating system and can be tailored to contain just the needed services for the app, using fewer system resources	Run the complete operating system including the kernel, which requires more system resources (CPU, memory, and storage)
Deployment	Deploy individual containers by using Docker via the command line; deploy multiple containers by using an orchestrator such as Azure Kubernetes Service	Deploy individual VMs by using Windows Admin Center or Hyper-V Manager; deploy multiple VMs by using PowerShell or System Center Virtual Machine Manager

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Compare Containers to Virtual Machines

The following table depicts the differences between containers and virtual machines:

Feature	Containers	Virtual Machines
Persistent storage	Use Azure Disks for local storage for a single node, or Azure Files (SMB shares) for storage shared by multiple nodes or servers	Use a virtual hard disk (VHD) for local storage for a single VM, or an SMB file share for storage shared by multiple servers
Fault tolerance	Recreate the running containers in case of a cluster node failure, by the orchestrator on another cluster node	Restarting a new server with operational VMs can fail the VMs over to another server in a cluster

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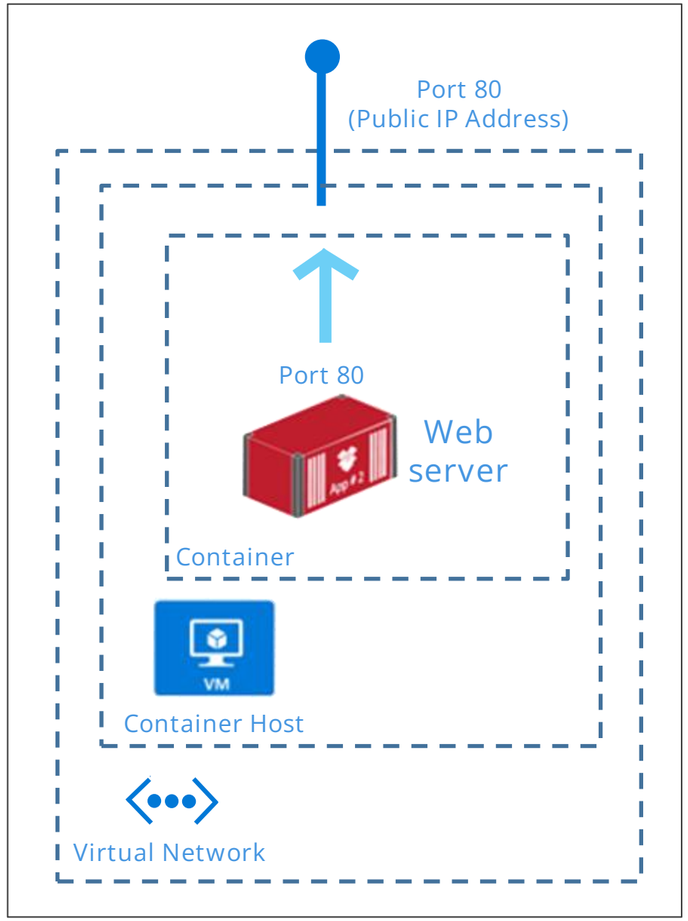
Explore Azure Container Instances Benefits

The fastest and simplest way to run containers in Azure is using Azure Container Instances (ACI), which do not need virtual machine management or higher-level services.



Benefits of Azure container instances:

- PaaS service
- Fast startup times
- Public IP connectivity and DNS name
- Isolation features
- Custom sizes
- Persistent storage
- Linux and Windows containers
- Co-scheduled groups
- Virtual network deployment

Source: Source: https://docs.microsoft.com/en-us/azure

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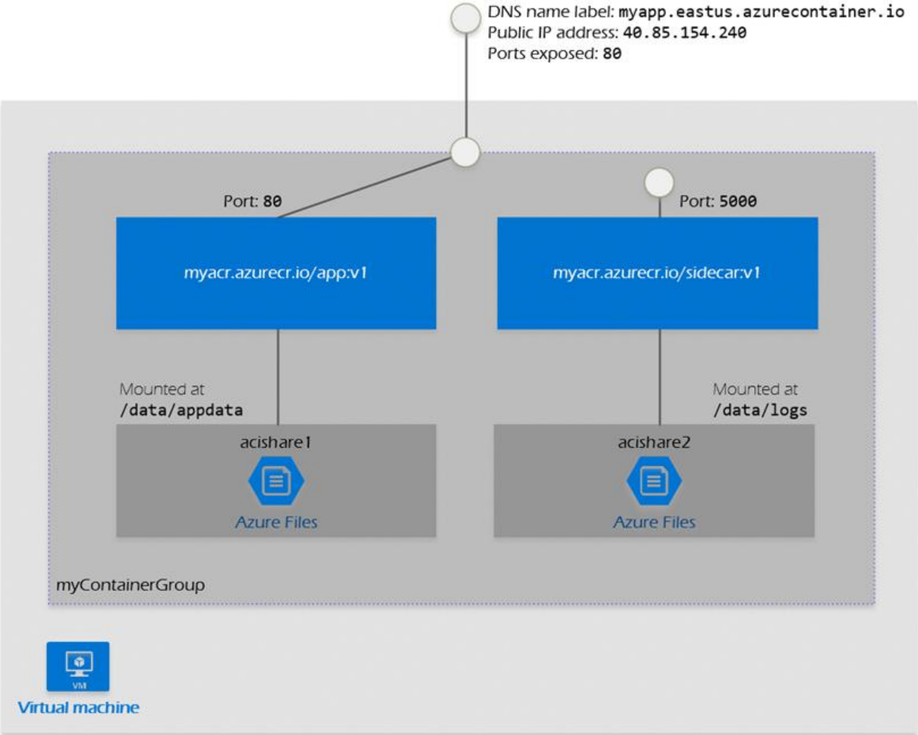
Implement Container Groups

It is a group of containers that are scheduled to run on the same host machine.

Features of container groups:

- Top-level resource in Azure container instances
- A collection of containers that get scheduled on the same host
- Containers in a group share a lifecycle, resources, local network, and storage volumes



Source: https://learn.microsoft.com/en-us/azure/container-instances/container-instances-container-groups

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Managing Container with Azure Container Apps

Duration: 10 Min.

Problem Statement:

You've been tasked to manage the container with Azure container apps.

UNASSISTED PRACTICE

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Assisted Practice: Guidelines

Steps to be followed:

1. Manage container with Azure container apps

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Configure Azure Container Instances

Duration: 10 Min.

Problem Statement:

You've been tasked to configure Azure container instances.

UNASSISTED PRACTICE

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Assisted Practice: Guidelines

Steps to be followed:

1. Configure Azure container instances

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Manage Sizing and Scaling for Containers

Azure container apps is a managed environment in Microsoft Azure that simplifies running containerized applications and microservices without the need for complex orchestration or infrastructure management.

Overview

Scaling: It is governed by scale limits, rules, and behavior.

Limits: It defines the range of replicas per revision, and scale rules are configured using HTTP, TCP, or custom protocols.

Rules: It defines when to add or remove replicas.

Behavior: It defines how rules and limits interact to make scaling decisions.

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Scale Rules

It is divided into three categories of triggers:

HTTP

It scales based on incoming HTTP requests, adjusting app instances automatically to web traffic.

TCP

It scales container app instances in Azure container apps but focuses on traffic using TCP.

Custom

It depends on CPU usage, memory usage, or compatible event-driven data sources.

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Azure Container Instances

- It offers on-demand provisioning of isolated containers.
- It serves as a versatile building block with flexible deployment options.
- It does not include built-in features such as scaling, load balancing, and certificate management; scaling to multiple container instances in ACI requires creating them individually.
- It is an excellent choice when your use case doesn't require the specific features provided by Azure container apps.

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Key Takeaways

- Azure Virtual Machines (VMs) are one of the several types of on-demand, scalable computing resources that Azure offers.
- Availability sets, availability zones, and virtual machine scale sets are the high-availability options for virtual machines.
- Azure VM extensions are small applications that provide post-deployment configuration and automation tasks on Azure VMs.
- Azure App Service is an HTTP-based service for hosting web applications.

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Creating a Virtual Machine Scale Set and Install Web Server on It

Duration: 15 Min.

Project agenda: To implement a virtual machine scale set with Arm templates and install IIS over those.

Description: The user has been given a project to create Virtual machine scale sets. Once the scale set has been created, we will be deploying the IIS webserver on these virtual machines present in the scale set.

Perform the following:

Create scale sets with the help of ARM templates and assign web server roles to these machines.

LESSON-END PROJECT

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